

Assignment-1

1. Write a program (WAP) to display "Hello World" on console display.

Code:-

```
ques8.cpp  ques11.cpp
1  //Assignment- 1
2  //Question -1
3  #include <iostream>
4  using namespace std;
5  int main() {
6      cout << "Hello World";
7      return 0;
8  }
```

Output:-

```
C:\Users\asus\Desktop\oops\  ×  +  v
Hello World
-----
Process exited after 0.2986 seconds with return value 0
Press any key to continue . . .
```

2. WAP to input an integer, decimal and character and display it.

Code:-

```
ques8.cpp  ques2.cpp
1  //Assignment- 1
2  //Question -2
3  #include <iostream>
4  using namespace std;
5  int main() {
6      int a;
7      float b;
8      char c;
9
10     cout << "Enter integer: ";
11     cin >> a;
12
13     cout << "Enter decimal: ";
14     cin >> b;
15
16     cout << "Enter character: ";
17     cin >> c;
18
19     cout << "Integer: " << a << endl;
20     cout << "Decimal: " << b << endl;
21     cout << "Character: " << c << endl;
22
23     return 0;
24 }
```

Output:-

```
C:\Users\asus\Desktop\oops\ × + v
Enter integer: 3
Enter decimal: 3.4
Enter character: A
Integer: 3
Decimal: 3.4
Character: A

-----
Process exited after 5.709 seconds with return value 0
Press any key to continue . . .
```

3. WAP to add, subtract, multiply, divide numbers.

Code:-

```
ques3.cpp
1 //Assignment- 1
2 //Question -3
3 #include <iostream>
4 using namespace std;
5 int main() {
6     float a, b;
7     cout << "Enter two numbers: ";
8     cin >> a >> b;
9
10    cout << "Addition: " << a + b << endl;
11    cout << "Subtraction: " << a - b << endl;
12    cout << "Multiplication: " << a * b << endl;
13
14    if(b != 0)
15        cout << "Division: " << a / b << endl;
16    else
17        cout << "Division: Not possible (divide by 0)\n";
18
19    return 0;
20 }
```

Output:-

```
C:\Users\asus\Desktop\oops\ × + v
Enter two numbers: 3 4
Addition: 7
Subtraction: -1
Multiplication: 12
Division: 0.75

-----
Process exited after 2.009 seconds with return value 0
Press any key to continue . . .
```

4. Write a C++ program that will ask for a temperature in Celsius and display it in degree Fahrenheit. $[F=9C/5+32]$

Code:-

```
ques3.cpp  ques4.cpp
1  //Assignment- 1
2  //Question -4
3  #include <iostream>
4  using namespace std;
5  int main() {
6      float c, f;
7      cout << "Enter temperature in Celsius: ";
8      cin >> c;
9
10     f = (9 * c / 5) + 32;
11
12     cout << "Temperature in Fahrenheit: " << f;
13     return 0;
14 }
```

Output:-

```
C:\Users\asus\Desktop\oops\  ×  +  v
Enter temperature in Celsius: 45
Temperature in Fahrenheit: 113
-----
Process exited after 2.197 seconds with return value 0
Press any key to continue . . .
```

5. WAP to format console output using '\n', '\t', '\b', endl, setw within cout statement

Code:-

```
ques3.cpp  ques4.cpp  ques5.cpp
1  //Assignment- 1
2  //Question -5
3  #include <iostream>
4  #include <iomanip>
5  using namespace std;
6  int main() {
7      cout << "Hello\nWorld" << endl;
8      cout << "Hello\tWorld" << endl;
9      cout << "Hello\b World" << endl;
10
11     cout << setw(10) << "C++" << endl;
12     cout << setw(10) << 123 << endl;
13
14     return 0;
15 }
```

Output:-

```
C:\Users\asus\Desktop\oops\  X  +  v
Hello
World
Hello World
Hello World
      C++
      123

-----
Process exited after 0.2729 seconds with return value 0
Press any key to continue . . .
```

6. WAP to utilize assignment += and -= type operators.

Code:-

```
ques3.cpp  ques4.cpp  ques5.cpp  ques66.cpp  ques6.cpp
1  //Assignment- 1
2  //Question -6
3  #include <iostream>
4  using namespace std;
5  int main() {
6      int a = 10;
7
8      a += 5;    // a = a + 5
9      cout << "After += : " << a << endl;
10
11     a -= 3;    // a = a - 3
12     cout << "After -= : " << a << endl;
13
14     return 0;
15 }
16
17
```

Output:-

```
C:\Users\asus\Desktop\oops\  X  +  v
After += : 15
After -= : 12

-----
Process exited after 0.3006 seconds with return value 0
Press any key to continue . . .
```

7. WAP to swap two numbers using bitwise operator.

Code:-

```

ques3.cpp  ques4.cpp  ques5.cpp  [*] Untitled9  ques6.cpp
1  //Assignment- 1
2  //Question -7
3  #include <iostream>
4  using namespace std;
5  int main() {
6      int a = 5, b = 10;
7      a = a ^ b;
8      b = a ^ b;
9      a = a ^ b;
10     cout << "After swap: a=" << a << " b=" << b;
11     return 0;
12 }

```

Output:-

```

C:\Users\asus\Desktop\oops\  ×  +  v
After swap: a=10 b=5
-----
Process exited after 0.2712 seconds with return value 0
Press any key to continue . . .

```

8. Write a program to solve the following problem A library charges a fine for every book return late. For first 5 days the fine is 50 paise, for 6-10 days fine is one rupee and above 10 days fine is 5 rupees. If you return the book after 30 days your membership will be cancelled .WAP to accept no. of days the member is late to return the book and display the fine or appropriate message.

Code:-

```

ques6.cpp  ques8.cpp
1  //Assignment- 1
2  //Question -8
3  #include <iostream>
4  using namespace std;
5  int main() {
6      int days;
7      cout << "Enter number of late days: ";
8      cin >> days;
9      if(days > 30) {
10         cout << "Membership Cancelled";
11     }
12     else if(days > 10) {
13         cout << "Fine: Rs. 5 per day";
14     }
15     else if(days > 5) {
16         cout << "Fine: Rs. 1 per day";
17     }
18     else if(days >= 1) {
19         cout << "Fine: 50 paise per day";
20     }
21     else {
22         cout << "No fine";
23     }
24 }

```

Output:-

```

C:\Users\asus\Desktop\oops\ x + v
Enter number of late days: 4
Fine: 50 paise per day
-----
Process exited after 1.748 seconds with return value 0
Press any key to continue . . .

```

9. WAP to display arithmetic operations using switch-case statement.

Code:-

```

ques6.cpp ques8.cpp ques9.cpp
1 //Assignment- 1
2 //Question -9
3 #include <iostream>
4 using namespace std;
5 int main() {
6     int a, b, choice;
7     cout << "Enter two numbers: ";
8     cin >> a >> b;
9     cout << "1.Add 2.Subtract 3.Multiply 4.Divide\n";
10    cout << "Enter choice: ";
11    cin >> choice;
12    switch(choice) {
13        case 1: cout << "Sum: " << a + b; break;
14        case 2: cout << "Difference: " << a - b; break;
15        case 3: cout << "Product: " << a * b; break;
16        case 4:
17            if(b != 0)
18                cout << "Division: " << a / b;
19            else
20                cout << "Divide by zero error";
21            break;
22        default:
23            cout << "Invalid choice";
24    }
25 }

```

Output:-

```

C:\Users\asus\Desktop\oops\ x + v
Enter two numbers: 3 4
1.Add 2.Subtract 3.Multiply 4.Divide
Enter choice: 3
Product: 12
-----
Process exited after 6.5 seconds with return value 0
Press any key to continue . . .

```

10. WAP to check whether the given number is even or odd (By using if-else and conditional operator).

Code:-

```
ques6.cpp  ques8.cpp  [*] ques9.cpp  ques10.cpp
1  //Assignment- 1
2  //Question -10
3  #include <iostream>
4  using namespace std;
5  int main() {
6      int n;
7      cout << "Enter number: ";
8      cin >> n;
9      if(n % 2 == 0)
10         cout << "Even\n";
11     else
12         cout << "Odd\n";
13     (n % 2 == 0) ? cout << "Even" : cout << "Odd";
14
15     return 0;
16 }
```

Output:-

```
C:\Users\asus\Desktop\oops\  ×  +  v
Enter number: 5
Odd
Odd
-----
Process exited after 3.636 seconds with return value 0
Press any key to continue . . .
```

